ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 6

MATHEMATICS

PAPER 1 · PRACTICE A

MARKS: 70 **TIME: 1½ hours**

An end-of-year test on Grade 6 Mathematics: whole numbers up to 9 digits, prime numbers, factors and multiples, decimals and percentages, number patterns, the four operations, common fractions, angles, 3D objects and nets, volume, perimeter and area, symmetry, measurement and time, a pie chart and probability.

Written by Academic Central for practice, following the CAPS curriculum for Grade 6. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL your working. You can get marks for working even if the answer is wrong.
4. You may NOT use a calculator.
5. Write neatly.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 What is the value of the digit 8 in 3 864 125?

A 8 000 B 80 000 C 800 000 D 8 000 000

1.2 Which number is a prime number?

A 9 B 15 C 21 D 23

1.3 0,7 written as a common fraction is …

A $^{7}/_{100}$ B $^{7}/_{10}$ C $^{1}/_{7}$ D $^{70}/_{10}$

1.4 25% of 80 = …

A 20 B 25 C 40 D 60

1.5 Which number is the biggest?

A 0,5 B 0,45 C 0,09 D 0,405

1.6 A triangle with all three sides equal is …

A an isosceles triangle. B a scalene triangle.
C an equilateral triangle. D a right-angled triangle.

1.7 When a shape is turned around a point, it is called a …

A reflection. B rotation. C translation. D enlargement.

1.8 4,5 cm = … mm

A 4,5 B 45 C 450 D 4 500

1.9 The lowest common multiple (LCM) of 4 and 6 is …

A 2 B 12 C 24 D 10

1.10 You roll an ordinary dice with the numbers 1 to 6. The chance of rolling a 7 is …

A certain. B likely. C unlikely. D impossible.

[10]

QUESTION 2: NUMBERS AND PATTERNS

2.1 Write 4 052 300 in words. (1)

2.2 Round 7 486 529 off to the nearest 100 000. (1)

2.3 Write down ALL the factors of 18. (2)

2.4 Write down the first THREE multiples of 9. (1)

2.5 Look at the number line in FIGURE 1. Which decimal numbers are at A and at B? (2)

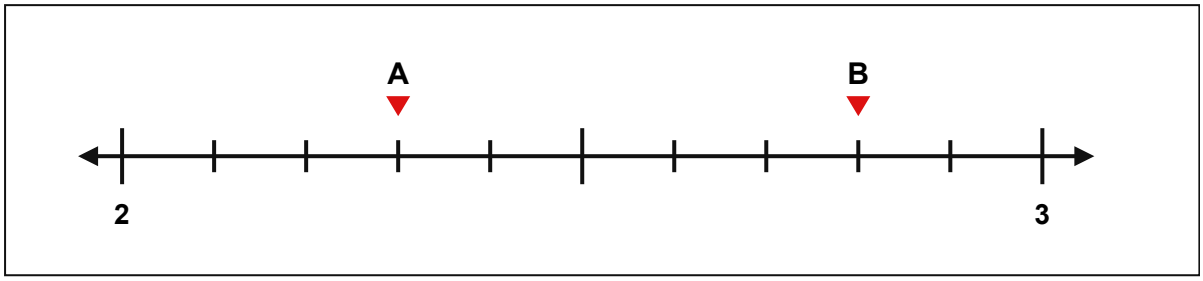


FIGURE 1: A number line

2.6 Write these numbers from the smallest to the biggest: 0,6; 0,06; 0,66; 6,0 (1)

2.7 Write $^{3}/_{4}$ as a decimal fraction and as a percentage. (2)

2.8 Look at the pattern: 2; 6; 18; 54; … Write the next number and the rule. (2)

[12]

QUESTION 3: CALCULATIONS

Show your working for each question.

3.1 45 678 + 128 954 (2)

3.2 300 000 − 145 387 (2)

3.3 2 345 × 23 (3)

3.4 9 360 ÷ 24 (3)

3.5 4,25 + 3,8 (1)

3.6 10 − 3,65 (1)

3.7 A school has R12 500 to spend on sport. It buys 15 soccer balls at R245 each. How much money is left? (3)

3.8 6 + 4 × 5 (1)

[16]

QUESTION 4: COMMON FRACTIONS

4.1 $^{1}/_{2} + {}^{1}/_{4}$ (2)

4.2 $^{5}/_{6} - {}^{1}/_{3}$ (2)

4.3 Find $^{2}/_{3}$ of 45. (2)

4.4 Write $^{12}/_{16}$ in its simplest form. (1)

4.5 Write $2\ {}^{1}/_{3}$ as an improper fraction. (1)

[8]

QUESTION 5: SPACE AND SHAPE

5.1 Write the type of each angle: acute, right, obtuse or reflex.

5.1.1 35° (1)

5.1.2 90° (1)

5.1.3 140° (1)

5.1.4 210° (1)

5.2 FIGURE 2 shows the net of a 3D object.

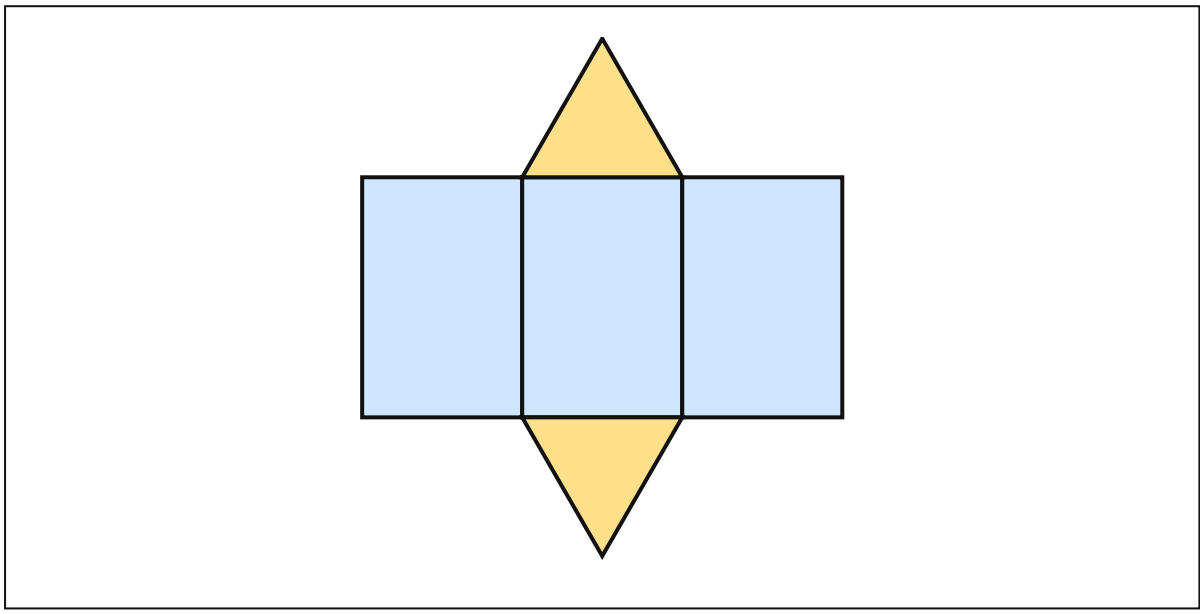


FIGURE 2: A net

5.2.1 Name the 3D object that this net folds into. (1)

5.2.2 How many faces does this object have? (1)

5.3 FIGURE 3 shows a stack of 1 cm cubes.

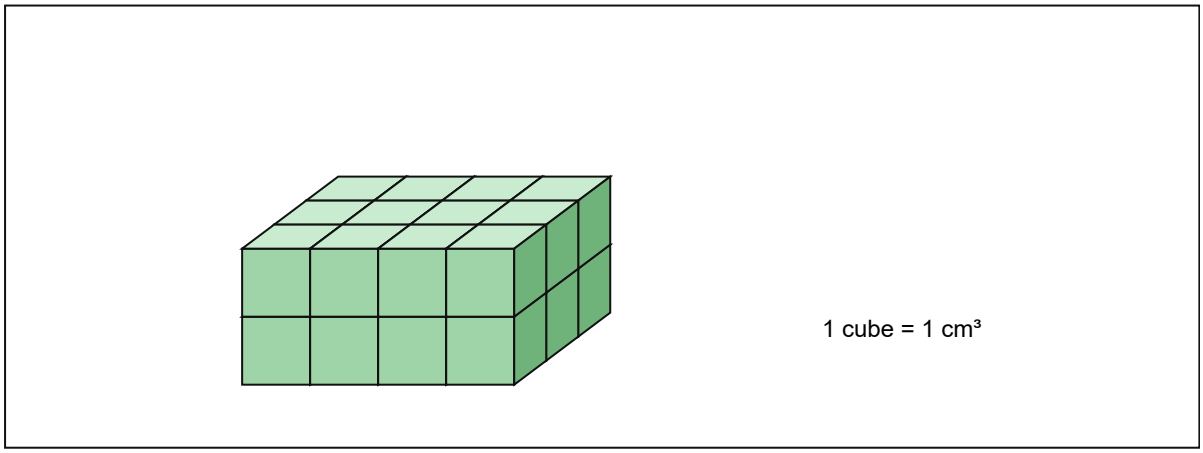


FIGURE 3: A stack of cubes

5.3.1 What is the volume of the stack? (1)

5.3.2 How many more layers of cubes must be added to make a volume of 48 cm³? (1)

5.4 Calculate:

5.4.1 the perimeter of a square with sides of 12 cm (1)

5.4.2 the area of a rectangle that is 9 m long and 7 m wide (2)

5.5 How many lines of symmetry does an equilateral triangle have? (1)

[12]

QUESTION 6: MEASUREMENT, DATA AND PROBABILITY

6.1 Convert:

6.1.1 3,2 kg to grams (1)

6.1.2 2 500 ml to litres (1)

6.2 A bus leaves Pretoria at 09:45 and arrives in Polokwane at 13:20. How long does the trip take? (2)

6.3 The temperature was 4 °C early in the morning and 21 °C in the afternoon. By how many degrees did the temperature rise? (1)

6.4 How many days are there in 3 weeks and 4 days? (1)

6.5 40 learners chose their favourite sport. The pie chart in FIGURE 4 shows the results.

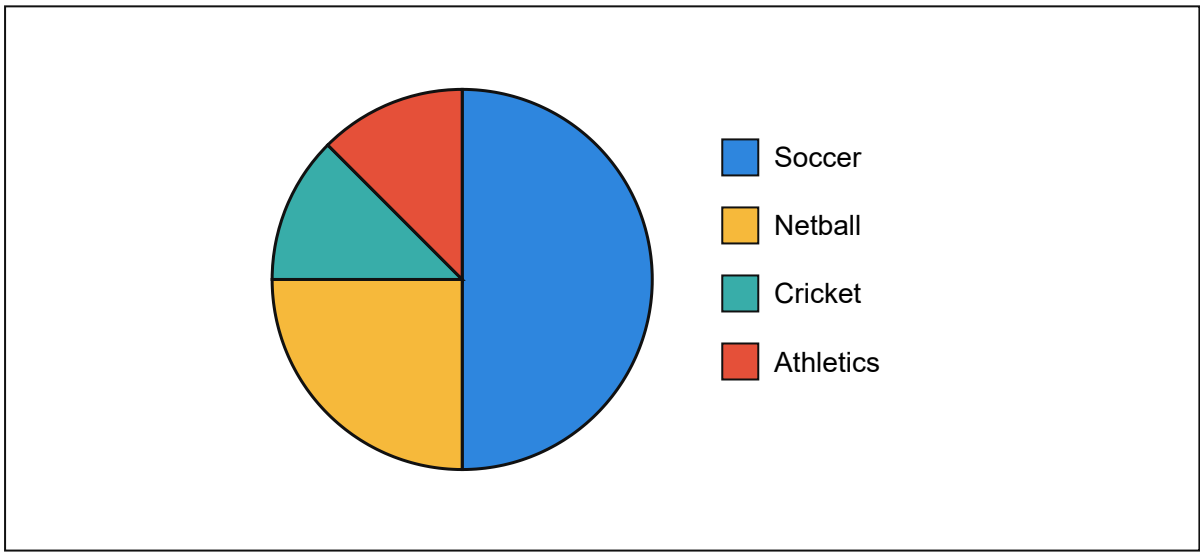


FIGURE 4: Our favourite sport

6.5.1 Which sport is the most popular? (1)

6.5.2 How many learners chose netball? Show your working. (2)

6.5.3 What fraction of the learners chose athletics? (1)

6.6 A spinner has 8 equal parts: 5 are blue, 2 are red and 1 is yellow.

6.6.1 On which colour is the spinner most likely to stop? (1)

6.6.2 Write the chance of stopping on red as a fraction. (1)

[12]

TOTAL: 70

END OF QUESTION PAPER